



Python Programming Fundamentals

VS Code Features

Programming Fundamentals Team

SETU

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Outline

1. VS Code Interface Overview
2. Essential Extensions for Python
3. Keyboard Shortcuts
4. IntelliSense
5. The Integrated Terminal
6. Virtual Environments
7. Multi-Cursor Editing



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1. VS Code Interface Overview

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Key areas of the VS Code interface:

Area	Purpose
Activity Bar (left)	Switch between Explorer, Search, Git, Extensions, Debug
Side Bar	File tree, search results, source control
Editor Area	Open files — supports split views
Panel (bottom)	Terminal, Output, Problems, Debug Console
Status Bar	Python interpreter, line/column, Git branch



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Opening a folder: File > Open Folder — always work in a folder, not a single file.



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Recommended:

- **autopep8** or **Black Formatter** — auto-format your code
- **GitLens** — enhanced Git integration
- **indent-rainbow** — colour-codes indentation levels
- **Error Lens** — shows errors inline in the editor
- **Python Docstring Generator** — auto-generates docstring templates



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After installing Python extension: select your interpreter from the status bar or Ctrl+Shift+P > Python: Select Interpreter.



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3. Keyboard Shortcuts

Knowing shortcuts makes you dramatically faster.

Mac	Windows/Linux	Action
Cmd+P	Ctrl+P	Quick Open file
Cmd+Shift+P	Ctrl+Shift+P	Command Palette
Cmd+/ 	Ctrl+/ 	Toggle line comment
Alt+Up/Down	Alt+Up/Down	Move line up/down
Opt+Click	Alt+Click	Add cursor (multi-cursor)
Cmd+D	Ctrl+D	Select next occurrence
Cmd+Shift+K	Ctrl+Shift+K	Delete line
F5	F5	Run / Debug
Ctrl+backtick	Ctrl+backtick	Open terminal



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4. IntelliSense

IntelliSense provides intelligent code completion as you type.

Start typing and VS Code suggests completions:

```
name = "hello"
```

```
name. # → shows: upper, lower, strip, split, replace...
```

```
import math
```

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math. # → shows: pi, e, sqrt, floor, ceil...
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Features:

- **Auto-complete** — press Tab or Enter to accept
- **Parameter hints** — shows function signatures as you type
- **Hover info** — hover over any name to see its type and docstring
- **Go to definition** — F12 to jump to where something is defined
- **Find all references** — Shift+F12



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Tip: Ctrl+Space triggers suggestions manually if they don't appear automatically.



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5. The Integrated Terminal

VS Code has a built-in terminal — no need to switch to a separate terminal app.

```
# Open terminal: Ctrl+` (backtick)
```

```
# Run your Python file  
python3 myfile.py
```

```
# Install packages  
pip install requests
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# Check Python version  
python3 --version
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Multiple terminals: click + in the terminal panel to open additional terminals.

Terminal types: VS Code uses your system shell (zsh on Mac, PowerShell/cmd on Windows). You can change the default in settings.



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6. Virtual Environments

A **virtual environment** isolates your project's packages from the system Python.

```
# Create a virtual environment
```

```
python3 -m venv .venv
```

```
# Activate it (Mac/Linux)
```

```
source .venv/bin/activate
```

```
# Activate it (Windows)
```

```
.venv\Scripts\activate
```

```
# Install packages (only in this environment)
```

```
pip install requests
```

```
# Deactivate when done
```

```
deactivate
```



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VS Code tip: when you create a `.venv` folder in your workspace, VS Code usually detects it automatically and selects it as the Python interpreter. Check the status bar — it should show `.venv`.



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Why use virtual environments?

- Avoid version conflicts between projects
- Keep your system Python clean
- Makes projects reproducible (use `pip freeze > requirements.txt`)



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7. Multi-Cursor Editing

Multi-cursor lets you edit multiple places simultaneously.

Add cursors manually:

- Alt+Click to add a cursor at each click location

Select all occurrences:

- Ctrl+D / Cmd+D — select next occurrence of current word
- Ctrl+Shift+L / Cmd+Shift+L — select all occurrences

```
# Example: rename a variable in many places
# Click on 'old_name', press Ctrl+Shift+L,
# then type 'new_name' – all instances update at once
old_name = 10
print(old_name)
result = old_name * 2
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7. Multi-Cursor Editing

Column selection: Shift+Alt+drag to select a rectangular block of text across multiple lines — useful for aligning columns.



Thanks for Watching - Any questions?

